

## 11. Initial Project Planning — Product Backlog & Sprint Schedule

|                      |                                                     |
|----------------------|-----------------------------------------------------|
| <b>Date</b>          | 28 June 2026                                        |
| <b>Team ID</b>       | SWTID-2026-9775                                     |
| <b>Project Name</b>  | EduGenie – Google Gemini Powered Learning Assistant |
| <b>Maximum Marks</b> | 5 Marks                                             |

| Sprint   | Epic                    | Story # | User Story / Task                                                                                                                   | Points | Priority | Team Member                                       | Sprint Dates |
|----------|-------------------------|---------|-------------------------------------------------------------------------------------------------------------------------------------|--------|----------|---------------------------------------------------|--------------|
| Sprint-1 | Core Setup              | USN-1   | As a developer, I can scaffold the FastAPI app, config loader and Jinja2/static setup so the project has a working shell.           | 2      | High     | D. Naga Sri Vidhya J. Melamudi                    | Wk 1         |
| Sprint-1 | Core Setup              | USN-2   | As a developer, I can build the JSON data-store manager (utils/db_manager.py) and seed a default user.                              | 2      | High     | Y. Sannidhi<br>Y. Sannidhi                        | Wk 1         |
| Sprint-1 | Q&A Module              | USN-3   | As a student, I can ask a question and receive a structured Gemini-powered answer (answer, explanation, example, key points).       | 3      | High     | B. Tallapaneni                                    | Wk 1–2       |
| Sprint-1 | Q&A Module              | USN-4   | As a developer, I can fall back to a demo/mock response when no Gemini API key is configured.                                       | 2      | High     | B. Tallapaneni                                    | Wk 2         |
| Sprint-2 | Explanation Module      | USN-5   | As a student, I can get a concept explained locally via LaMini-Flan-T5, with automatic fallback to Gemini if the local model fails. | 3      | Medium   | S. Rajitha Annapureddy,<br>S. Rajitha Annapureddy | Wk 2–3       |
| Sprint-2 | Quiz Module             | USN-6   | As a student, I can generate an MCQ quiz (1–15 questions) from a topic or pasted text, with explanations per answer.                | 3      | High     | D. Naga Sri Vidhya J. Melamudi                    | Wk 3         |
| Sprint-2 | Quiz Module             | USN-7   | As a student, I can take the quiz in the UI with a timer, see my score, and review explanations (quiz.js).                          | 2      | Medium   | S. Rajitha Annapureddy,<br>S. Rajitha Annapureddy | Wk 3         |
| Sprint-3 | Summary & Learning Path | USN-8   | As a student, I can paste notes and get a summary with bullet points and keywords.                                                  | 2      | Medium   | Y. Sannidhi<br>Y. Sannidhi                        | Wk 4         |
| Sprint-3 | Summary & Learning Path | USN-9   | As a student, I can request a personalized learning roadmap                                                                         | 3      | Medium   | D. Naga Sri Vidhya J. Melamudi                    | Wk 4         |

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|----------|----------------------|---------|------------------------------------------------------------------------------------------------------------------|--------|----------|----------------------------|--------------|
|          |                      |         | with resources and a step-by-step plan for a topic.                                                              |        |          |                            |              |
| Sprint-3 | Testing & Deployment | USN-10  | As a developer, I can verify all routes and validation rules with an automated pytest suite (tests/test_api.py). | 2      | High     | B. Tallapaneni             | Wk 4-5       |
| Sprint-3 | Testing & Deployment | USN-11  | As a developer, I can containerize the app with a Dockerfile for consistent deployment.                          | 2      | Medium   | Y. Sannidhi<br>Y. Sannidhi | Wk 5         |